Measure Theoretic Probability
Lesson 27
Integral Properties, Measures from Integrals, Continuity of Measures

Kirschner Awonatey Amaanya

Department of Statistics and Actuarial Science, KNUST

Outline

1. Recap: the Lebesgue Integral
2. Fact 5: Null Sets Carry No Integral
3. Fact 6: Indicators Move Into the Integrand
4. Two More Results on Simple Functions
5. Preview: Monotone Convergence Theorem
6. Continuity of General Measures
7. Appendix: Proof of Fubini I'
8. Summary and Exercises

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Setting and Definitions

Setting

$(\Omega, \mathcal{F}, \mu)$ is a measure space. All functions today are **measurable** and **non-negative**. Ω need not be numerical.

Simple function

$s = \sum_{i=1}^{n} a_i I_{A_i}$, $a_i \geq 0$, $A_i \in \mathcal{F}$. For $A \in \mathcal{F}$,

$$\int_A s \, \mathrm{d}\mu = \sum_{i=1}^{n} a_i \, \mu(A_i \cap A).$$

Non-negative measurable f

$$\int_A f \, \mathrm{d}\mu = \sup\left\{ \int_A s \, \mathrm{d}\mu : \ s \text{ simple, measurable, } 0 \leq s \leq f \right\}.$$

Picture

Heights are measured as usual (on $\mathbb{R}$), but each base is measured by μ. For $\mu = \mathbb{P}$: $\mathbb{E}[X] = \int_\Omega X \, \mathrm{d}\mathbb{P}$.

Facts 1 to 4

For $f, g \geq 0$ measurable and $A, B \in \mathcal{F}$

1. $0 \leq f \leq g \Rightarrow \int_A f \, \mathrm{d}\mu \leq \int_A g \, \mathrm{d}\mu$.
2. $A \subseteq B \Rightarrow \int_A f \, \mathrm{d}\mu \leq \int_B f \, \mathrm{d}\mu$.
3. $c \geq 0 \Rightarrow \int_A cf \, \mathrm{d}\mu = c \int_A f \, \mathrm{d}\mu$.
4. $f = 0$ on $A \Rightarrow \int_A f \, \mathrm{d}\mu = 0$.

Common proof pattern

Compare the simple functions under each side, then use that a supremum is the **least** upper bound: if U is an upper bound of a set, then sup $\leq U$. We use this pattern again today.

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Fact 5: $\mu(A) = 0 \Rightarrow \int_A f \, \mathrm{d}\mu = 0$

Idea

The base of every rectangle over A has measure 0, so no area accumulates, whatever the heights are.

Proof

It suffices to show $\int_A s \, \mathrm{d}\mu = 0$ for every simple s with $0 \le s \le f$ (the supremum of zeros is 0). Write $s = \sum_i a_i I_{A_i}$. Since $A_i \cap A \subseteq A$, monotonicity of μ gives

$$0 \le \mu(A_i \cap A) \le \mu(A) = 0.$$

Hence

$$\int_A s \, \mathrm{d}\mu = \sum_i a_i \, \mu(A_i \cap A) = 0. \qquad \square$$

Contrast with Fact 4

Fact 4: $f = 0$ on A (any $\mu(A)$). Fact 5: $\mu(A) = 0$ (any f). Either way the integral over A is 0. (If $a_i = \infty$ were allowed, we would use $0 \cdot \infty = 0$; simple functions here have finite coefficients.)

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Fact 6: Statement and Idea

Fact 6

For $f \geq 0$ measurable and $A \in \mathcal{F}$,

$$\int_A f \, \mathrm{d}\mu = \int_\Omega I_A f \, \mathrm{d}\mu.$$

Idea

Integrating over A is the same as integrating over everything after switching f off outside A. This is used constantly, and it lets us always work with integrals over Ω.

Plan

- Case 1: f simple (direct computation).
- Case 2: general f, two inequalities ($\leq$ and $\geq$), each via the supremum pattern.

Fact 6, Case 1: f Simple

Computation

Let $f = \sum_i a_i I_{A_i}$. A product of indicators is the indicator of the intersection, so

$$I_A f = \sum_i a_i \, I_{A_i \cap A},$$

which is again simple (the sets may no longer form a partition, but that is fine). Therefore

$$\int_\Omega I_A f \, \mathrm{d}\mu = \sum_i a_i \, \mu(A_i \cap A) = \int_A f \, \mathrm{d}\mu. \qquad \square$$

Numerical check

$\Omega = [0,1]$, λ Lebesgue, $f = 2I_{[0,\frac{1}{2})} + 4I_{[\frac{1}{2},1]}$, $A = [\frac{1}{4}, \frac{3}{4}]$. Then $I_A f = 2I_{[\frac{1}{4},\frac{1}{2})} + 4I_{[\frac{1}{2},\frac{3}{4}]}$, so

$$\int_\Omega I_A f \, \mathrm{d}\lambda = 2 \cdot \tfrac{1}{4} + 4 \cdot \tfrac{1}{4} = \tfrac{3}{2} = 2 \cdot \lambda\big([0, \tfrac{1}{2}) \cap A\big) + 4 \cdot \lambda\big([\tfrac{1}{2}, 1] \cap A\big).$$

Fact 6, Case 2: General f, Part (a): $\leq$

Claim

$$\int_A f \, \mathrm{d}\mu \leq \int_\Omega I_A f \, \mathrm{d}\mu.$$

Proof

Let s be simple with $0 \leq s \leq f$. Multiplying by $I_A \geq 0$ keeps the inequalities:

$$0 \leq I_A s \leq I_A f.$$

By Case 1 and Fact 1 (applied to $I_A s \leq I_A f$),

$$\int_A s \, \mathrm{d}\mu = \int_\Omega I_A s \, \mathrm{d}\mu \;\leq\; \int_\Omega I_A f \, \mathrm{d}\mu.$$

So $\int_\Omega I_A f \, \mathrm{d}\mu$ is an upper bound of all the numbers $\int_A s \, \mathrm{d}\mu$. The supremum is the least upper bound, so

$$\int_A f \, \mathrm{d}\mu = \sup_s \int_A s \, \mathrm{d}\mu \leq \int_\Omega I_A f \, \mathrm{d}\mu. \qquad \square$$

Fact 6, Case 2: Part (b): $\geq$

Claim

$$\int_\Omega I_A f \, \mathrm{d}\mu \leq \int_A f \, \mathrm{d}\mu.$$

Take any simple s with $0 \leq s \leq I_A f$

- $I_A f \leq f$ (the indicator is 0 or 1), so also $0 \leq s \leq f$.
- s vanishes off A: for $\omega \notin A$, $0 \leq s(\omega) \leq I_A(\omega) f(\omega) = 0$.
- Hence $s = I_A s$ (both sides equal s on A and 0 off A).

Conclude

By Case 1, then Fact 1 ($s \leq f$):

$$\int_\Omega s \, \mathrm{d}\mu = \int_\Omega I_A s \, \mathrm{d}\mu = \int_A s \, \mathrm{d}\mu \leq \int_A f \, \mathrm{d}\mu.$$

Thus $\int_A f \, \mathrm{d}\mu$ is an upper bound for all $\int_\Omega s \, \mathrm{d}\mu$ with $0 \leq s \leq I_A f$, so $\int_\Omega I_A f \, \mathrm{d}\mu \leq \int_A f \, \mathrm{d}\mu$. Together with (a), equality holds. □

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Switching Double Sums ("Fubini I")

Theorem

If $a_{ij} \geq 0$ for all $i, j \geq 1$, then

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

(both sides may be $+\infty$).

Why it matters

Non-negativity prevents cancellation, so the order of summation cannot change the total. We use this in the next frames, and it is the same idea behind the monotone convergence theorem.

Warning

For terms of mixed sign, the order can matter. Non-negativity is essential.

Additivity on Simple Functions

Fact 7

For non-negative simple s_1, s_2 and $A \in \mathcal{F}$:

$$\int_A (s_1 + s_2)\,\mathrm{d}\mu = \int_A s_1\,\mathrm{d}\mu + \int_A s_2\,\mathrm{d}\mu.$$

Proof sketch (left as an exercise in the lecture)

Write $s_1 = \sum_i a_i I_{A_i}$ and $s_2 = \sum_j b_j I_{B_j}$ with $\{A_i\}, \{B_j\}$ partitions of Ω. Then

$$s_1 + s_2 = \sum_{i,j} (a_i + b_j) I_{A_i \cap B_j}$$

is simple, and

$$\begin{aligned}\int_A (s_1 + s_2)\,\mathrm{d}\mu &= \sum_{i,j} (a_i + b_j)\mu(A_i \cap B_j \cap A) \\ &= \sum_i a_i \mu(A_i \cap A) + \sum_j b_j \mu(B_j \cap A),\end{aligned}$$

by finite additivity of μ ($A_i \cap A = \bigsqcup_j A_i \cap B_j \cap A$, and similarly for $B_j \cap A$). □

Theorem: $A \mapsto \int_A s \, \mathrm{d}\mu$ Is a Measure

Theorem

Let $s = \sum_{i=1}^{m} b_i I_{B_i}$ be non-negative simple. Define

$$\nu(A) = \int_A s \, \mathrm{d}\mu, \qquad A \in \mathcal{F}.$$

Then ν is a measure on $(\Omega, \mathcal{F})$.

We check three things

1. $\nu(A) \geq 0$.
2. $\nu(\varnothing) = 0$.
3. Countable additivity.

Meaning

An integral of a fixed function gives a **new way of measuring sets**. If $\mu = \mathbb{P}$ and s is a density-like weight, $\nu(A)$ is "the s-weighted probability of A".

Proof: Non-Negativity and $\nu(\varnothing) = 0$

1. $\nu(A) \geq 0$

$\nu(A) = \sum_i b_i \, \mu(B_i \cap A)$, with every $b_i \geq 0$ and every $\mu(\cdot) \geq 0$. So $\nu(A) \geq 0$.

2. $\nu(\varnothing) = 0$

$\mu(\varnothing) = 0$, so Fact 5 gives $\nu(\varnothing) = \int_{\varnothing} s \, \mathrm{d}\mu = 0$. Directly: $\mu(B_i \cap \varnothing) = 0$ for each i.

Remark

These are the "first line" conditions of the definition of a measure: values in $[0, \infty]$ and $\nu(\varnothing) = 0$.

Proof: Countable Additivity

Let $A_1, A_2, \dots \in \mathcal{F}$ be pairwise disjoint, $A = \bigsqcup_n A_n$

$$\nu(A) = \sum_{i=1}^{m} b_i \, \mu(B_i \cap A), \qquad B_i \cap A = \bigsqcup_{n=1}^{\infty} (B_i \cap A_n).$$

The sets $B_i \cap A_n$ are disjoint in n, so countable additivity of μ gives

$$\nu(A) = \sum_{i=1}^{m} \sum_{n=1}^{\infty} b_i \, \mu(B_i \cap A_n).$$

Switch the order of summation

All terms are ≥ 0, and the sum over i is finite, so the order may be exchanged (Fubini I'):

$$\nu(A) = \sum_{n=1}^{\infty} \sum_{i=1}^{m} b_i \, \mu(B_i \cap A_n) = \sum_{n=1}^{\infty} \int_{A_n} s \, \mathrm{d}\mu = \sum_{n=1}^{\infty} \nu(A_n). \qquad \square$$

Finite or not?

$\nu(\Omega) = \int_\Omega s \, \mathrm{d}\mu$. If s is integrable, ν is a **finite measure**. Otherwise $\nu(\Omega) = \infty$ is possible, because $\mu(B_i)$ may be infinite even though the sum has finitely many terms.

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Lebesgue's Monotone Convergence Theorem (Statement)

Theorem (to be proved next lesson)

Let $f_n : \Omega \to [0, \infty)$ be measurable with

$$0 \leq f_1(\omega) \leq f_2(\omega) \leq \cdots \quad \text{for all } \omega, \qquad f_n(\omega) \to f(\omega) \in [0, \infty].$$

Then (i) f is measurable, and (ii)

$$\lim_{n \to \infty} \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Meaning

For an increasing sequence of non-negative functions, the limit can be moved **inside** the integral. The same holds over any set $A \in \mathcal{F}$.

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What Changes Without $\mathbb{P}(\Omega) = 1$

Probability measures

Continuity from above was proved by passing to complements: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

General measures

$\mu(A) + \mu(A^c) = \mu(\Omega)$, which may be ∞. Then $\mu(A^c) = \mu(\Omega) - \mu(A)$ is meaningless.

Statement for a general measure space

- **From below:** $A_n \uparrow A \Rightarrow \mu(A_n) \uparrow \mu(A)$ (no extra condition).
- **From above:** $A_1 \supseteq A_2 \supseteq \cdots$ with $A_n \downarrow A = \bigcap_n A_n$, **and** $\mu(A_1) < \infty \Rightarrow \mu(A_n) \downarrow \mu(A)$.

Weaker than finiteness of $\mu(\Omega)$

We only need $\mu(A_1) < \infty$ for the first set. This holds automatically if $\mu(\Omega) < \infty$.

Proof of Continuity from Below

Disjointify

Let $A_1 \subseteq A_2 \subseteq \cdots$, $A = \bigcup_n A_n$. Put

$$B_1 = A_1, \qquad B_n = A_n \setminus A_{n-1} \ (n \geq 2).$$

The B_n are disjoint, $A_n = \bigsqcup_{k=1}^{n} B_k$ (a finite union), and $A = \bigsqcup_{n=1}^{\infty} B_n$.

Compute

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \lim_{M \to \infty} \sum_{n=1}^{M} \mu(B_n) = \lim_{M \to \infty} \mu(A_M).$$

The sequence $\mu(A_M)$ is non-decreasing by monotonicity ($A_M \subseteq A_{M+1}$), so it increases to $\mu(A)$. □

Picture

Think of nested rings: B_1 is the core, B_2 the ring around it, B_3 the next ring. The rings are disjoint and together build A.

Proof of Continuity from Above

Assume $A_1 \supseteq A_2 \supseteq \cdots$, $A = \bigcap_n A_n$, $\mu(A_1) < \infty$

Consider the sets $C_n = A_1 \setminus A_n$ ($n \geq 1$). They increase: $C_n \subseteq C_{n+1}$, and

$$\bigcup_n C_n = A_1 \setminus \bigcap_n A_n = A_1 \setminus A.$$

By continuity from below, $\mu(C_n) \uparrow \mu(A_1 \setminus A)$.

Subtract (this is where $\mu(A_1) < \infty$ is used)

Since $A_n \subseteq A_1$ and $A \subseteq A_1$, with all measures finite,

$$\mu(C_n) = \mu(A_1) - \mu(A_n), \qquad \mu(A_1 \setminus A) = \mu(A_1) - \mu(A).$$

Hence $\mu(A_1) - \mu(A_n) \uparrow \mu(A_1) - \mu(A)$, i.e.

$$\mu(A_n) \downarrow \mu(A). \qquad \square$$

Counterexample without finiteness

Lebesgue measure on $\mathbb{R}$, $A_n = [n, \infty)$. Then $A_n \downarrow \varnothing$ but $\lambda(A_n) = \infty$ for all n, so it does **not** converge to $\lambda(\varnothing) = 0$.

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Proof of Fubini I' (Setup)

Notation for $a_{ij} \geq 0$

$$R_i = \sum_{j=1}^{\infty} a_{ij}, \quad C_j = \sum_{i=1}^{\infty} a_{ij}, \quad R = \sum_{i=1}^{\infty} R_i, \quad C = \sum_{j=1}^{\infty} C_j.$$

We must show $R = C$. Define

$$D = \sup\Big\{ \sum_{(i,j)\in S} a_{ij} : \ S \subseteq \mathbb{N} \times \mathbb{N} \text{ finite} \Big\}.$$

We show $R = D$; the argument for $C = D$ is symmetric. Then $R = C$.

Idea

D is "add up any finite chunk of the table, take the best". Both row-first and column-first sums should capture exactly this.

Proof of Fubini I': $R = D$

$R \leq D$

For integers N, M, the set $S = \{1..N\} \times \{1..M\}$ is finite, so

$$\sum_{i=1}^{N} \sum_{j=1}^{M} a_{ij} \leq D.$$

Let $M \to \infty$: $\sum_{i=1}^{N} R_i \leq D$. Let $N \to \infty$: $R \leq D$.

$D \leq R$

Let S be any finite subset of $\mathbb{N} \times \mathbb{N}$. Choose N, M with $S \subseteq \{1..N\} \times \{1..M\}$. Terms are ≥ 0, so enlarging the index set (and letting $M \to \infty$) only increases the sum:

$$\sum_{(i,j) \in S} a_{ij} \leq \sum_{i=1}^{N} \sum_{j=1}^{M} a_{ij} \leq \sum_{i=1}^{N} R_i \leq R.$$

So R is an upper bound for all finite partial sums, and $D \leq R$.

Conclusion

$R = D$, and symmetrically $C = D$, hence $R = C$. □

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Summary

Key results

Concept	Content
Fact 5	$\mu(A) = 0 \Rightarrow \int_A f \, \mathrm{d}\mu = 0$
Fact 6	$\int_A f \, \mathrm{d}\mu = \int_\Omega I_A f \, \mathrm{d}\mu$ (simple case, then $\leq$ and $\geq$ via suprema)
Fact 7	$\int_A (s_1 + s_2) = \int_A s_1 + \int_A s_2$ for simple s_i
Measure from integral	$\nu(A) = \int_A s \, \mathrm{d}\mu$ is a measure; finite iff s integrable
Fubini I'	non-negative double series can be summed in either order
Continuity from below	$A_n \uparrow A \Rightarrow \mu(A_n) \uparrow \mu(A)$
Continuity from above	$A_n \downarrow A$, $\mu(A_1) < \infty \Rightarrow \mu(A_n) \downarrow \mu(A)$
MCT (preview)	$0 \leq f_n \uparrow f \Rightarrow \int f_n \uparrow \int f$

Exercises

1. Let μ be Lebesgue measure on $[0,1]$, $f = 3I_{[0,\frac{1}{3})} + 1I_{[\frac{1}{3},1]}$, $A = [\frac{1}{6}, \frac{1}{2}]$. Compute $\int_A f \, \mathrm{d}\mu$ and $\int_\Omega I_A f \, \mathrm{d}\mu$.
2. Let μ be the counting measure on $\mathbb{N}$ and $f(n) = 1/n^2$. Compute $\int_{\mathbb{N}} f \, \mathrm{d}\mu$ as a series. What does Fubini I' say about $\int_{\mathbb{N}} f \, \mathrm{d}\mu$ if you split $\mathbb{N}$ into evens and odds?
3. Show that Fact 5 and Fact 4 can both apply, and give an example where only one applies.
4. Prove Fact 7 in full, including why $s_1 + s_2$ is simple.
5. Take $\mu = \lambda$ on $[0,1]$ and $s = 2I_{[0,\frac{1}{2})}$. Compute $\nu([0,1])$ and $\nu([\frac{1}{4}, \frac{3}{4}])$ for $\nu(A) = \int_A s \, \mathrm{d}\lambda$.
6. Show a series with mixed signs where interchanging the order of a double sum changes the value. Which hypothesis of Fubini I' fails?
7. Give an example with $A_n \downarrow \varnothing$ but $\mu(A_n) \not\to 0$. Which hypothesis fails?
8. Where does the proof of Fact 6(b) use that s is simple?